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System Identification Report: Motor Modelling

1. System Selection and Approval

The system we chose is a motor modelling system. It has three types of motors: DC motor, stepper motor and servo motor.

This system is in Category E(Digital Twins / Electromechanical Systems). We picked it to study how electrical and mechanical parts work together. The system is approved for modelling, simulation and validation by our faculty.

2. Real-World Application

These motors are used in things like robotics, automation and control systems:

- DC motors are used for things that need to keep rotating, like fans and conveyors.
- Stepper motors are used for movements like in 3D printers and CNC machines.
- Servo motors are used in systems that need feedback like robot arms and drones.

This project is about modeling and comparing how these motors behave under conditions.

3. System Definition

1. Inputs

- Voltage applied: $V(t)$
- Load torque disturbance: $T_L(t)$
- For stepper motor: voltages $V_A(t)$ $V_B(t)$
- For servo motor: reference position $\theta_{ref}(t)$

2. Outputs

- Angular position: $\theta(t)$
- Angular velocity: $\omega(t)$
- Step position for stepper motor
- Tracking response for servo motor

3. State Variables

- DC Motor:

$$X_1 = \omega(t), \quad x_2 = i(t)$$

- Stepper Motor:

$$x_1 = \theta(t), \quad x_2 = \dot{\theta}(t), \quad x_3 = iA(t), \quad x_4 = iB(t)$$

- Servo Motor:

$$X_1 = \theta(t), \quad x_2 = \dot{\theta}(t), \quad x_3 = i(t)$$

4. Key Parameters

- J: Moment of inertia
- B: Damping or friction coefficient
- R: Resistance
- L: Inductance
- K_t : Torque constant
- K_e : EMF constant
- K_p, K_d : Controller gains for servo motor

4. Degrees of Freedom (DOF)

The DOF is 1 for all three motors.

Justification:

Each motor has one kind of motion which is the rotation of its shaft.

All system dynamics are described using one angular coordinate θ .

5. Key Assumptions

- Motors are treated as bodies.
- Magnetic effects are simple and constant.
- Friction is like viscous and changes, with velocity.
- Electrical parts like resistance and inductance are constant.
- There are no disturbances except load torque.
- Temperature effects are ignored.
- Stepper motor is modeled using a torque relation.
- Servo motor feedback is assumed to be ideal.
- The system operates within limits.

6. Free-Body Diagram

